Proving Angle Relationships — Notes

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

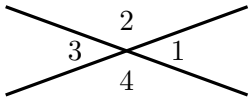
☞ *Video: Four facts you can use*

Key Idea

Linear Pair Postulate	a linear pair is _____ (adds to _____)
Vertical Angles Congruence Theorem	vertical angles are _____
Right Angles Congruence Theorem	all right angles are _____
Congruent Supplements / Complements	two angles supplementary to the _____ angle are congruent

☞ *Video: Why vertical angles are congruent*

Watch Me — Worked Example



$m\angle 1 + m\angle 2 = 180°$ _____

$m\angle 3 + m\angle 2 = 180°$ _____

$\angle 1 \cong \angle 3$ _____

☞ *Video: Use it to find angles*

Watch Me — Worked Example

Same picture: $m\angle 1 = 3x + 10,\ m\angle 3 = 5x - 30.$

$3x + 10 = 5x - 30$ vertical angles are _____

$40 = 2x,\ x =$ _____ solve

$m\angle 1 = 3(20) + 10 =$ _____

$m\angle 2 = 180 - 70 =$ _____ $\angle 1$ and $\angle 2$ are a _____

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☞ *Video: Watch out*

Common Mistake

Vertical angles are _____ from each other — they are equal. Angles _____ each other on a line are a linear pair — they add to $180°$, they are _____ equal.

☞ *Video: Your turn*

Your Turn

$\angle A$ and $\angle B$ are both supplementary to $\angle C$. $m\angle A = 2x + 15,\ m\angle B = 3x - 5$. Find x and $m\angle A$.

Congruent _____: $2x + 15 = 3x - 5 \Rightarrow x =$ _____, $m\angle A =$ _____

Notice

Remember: Linear pair: adds to $180°$. Vertical angles: congruent. All right angles are congruent. Supplementary (or complementary) to the **same** angle $\Rightarrow$ congruent.

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